

Covariate specification and model building

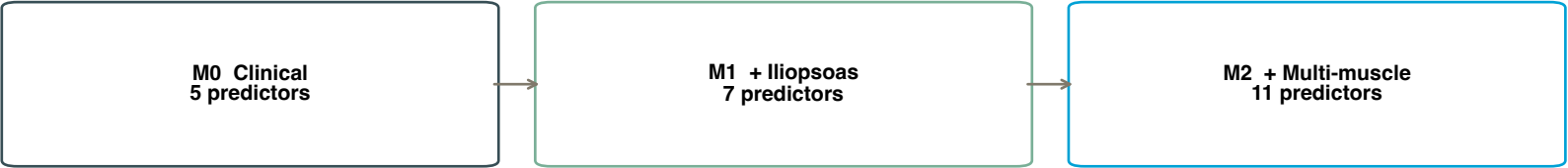
Candidate pool (PRE-SPECIFIED)

Clinical (adjustment set):
Age
Female sex
ASA class
No. of operated levels
Fusion vs decompression

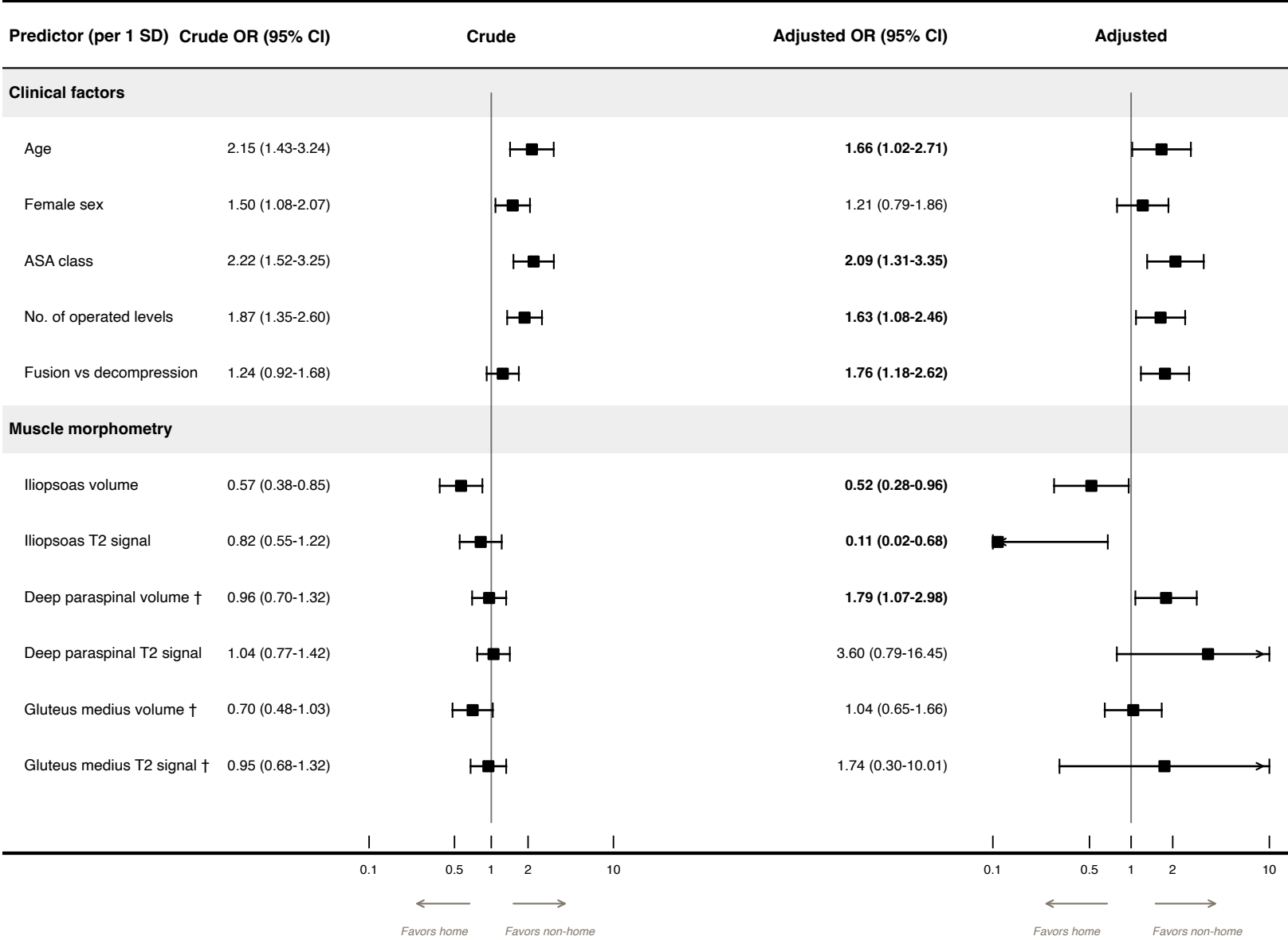
Imaging: iliopsoas · deep paraspinal ·
gluteus medius (volume + T2 signal)

Selection RULE

- No data-driven selection (no stepwise)
- Blocks entered by pre-planned nesting
 - Ridge (L2) shrinkage
 - EPV = 51/11 = 4.6



Crude and adjusted associations with non-home discharge (n=205; 51 events)



Crude = each predictor modeled alone. Adjusted = one ridge-penalized multivariable logistic model mutually adjusting for ALL listed covariates (age, female sex, ASA class, no. of operated levels, fusion) plus the three muscle groups (volume + T2 signal). Bold adjusted OR: P<.05. † crude-to-adjusted sign reversal (collinearity/suppression artifact; not interpreted). Squares = point estimate; whiskers = 95% CI; arrows = CI beyond axis range.